Comprehensive Guide to Supermesh Analysis

1. High-Level Overview

What is Mesh Analysis?

Before understanding a **Supermesh**, let us first define the building blocks:

- **Loop:** Any closed path in a circuit through which electric current can flow back to its starting point.
- **Mesh:** A specific type of loop that **does not contain any other loops inside it**. Think of a mesh as a single "windowpane" in a window frame.
- **Mesh Current:** An imaginary current circulating around a specific mesh in a defined direction (usually clockwise).

In standard **Mesh Analysis**, we apply **Kirchhoff's Voltage Law (KVL)** around every individual mesh in a circuit. KVL states:

$$\sum V = 0$$

(The algebraic sum of all potential differences around any closed loop must equal zero.)

Why do we need Supermesh Analysis?

When analyzing circuits, we frequently encounter **Current Sources** (devices that force a fixed, constant electric current to flow through a branch, regardless of the voltage across them).

This introduces a problem for basic KVL:

1. KVL requires us to sum the voltages across every component in a loop ($V = I \cdot R$).
2. The voltage across an ideal current source is **unknown** and cannot be calculated using Ohm's Law directly because an ideal current source dictates current, not resistance or voltage.

If a current source is located on the **shared boundary between two adjacent meshes**, standard KVL breaks down because we cannot write the voltage drop across that current source directly.

The Supermesh Solution

A **Supermesh** is formed by **virtually removing (open-circuiting) the shared current source** along with any resistor in series with it, and combining the two adjacent meshes into a single, larger perimeter loop.

Key points to remember:

1. **Combine the Meshes:** Merge Mesh 1 and Mesh 2 into one single outer loop (the "Supermesh") to bypass the current source completely.
2. **Apply KVL:** Apply Kirchhoff's Voltage Law around this new combined outer loop.
3. **Constraint Equation:** Create a second equation using the original current source to link the mesh currents (I_1 and I_2).

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2. Fundamental Rules & Conventions

To avoid common mistakes with signs, strictly adhere to these standard circuit conventions:

Sign Conventions for KVL

- **DC Voltage Sources:**
  - Moving from **Negative (-) to Positive (+)** terminal: Potential increases $\rightarrow +V$ (Voltage Gain).
  - Moving from **Positive (+) to Negative (-)** terminal: Potential decreases $\rightarrow -V$ (Voltage Drop).
- **Resistors:**
  - As current flows through a resistor R, it drops voltage in the direction of the current flow.
  - Moving in the **same direction** as your mesh current: $-I \cdot R$ (Voltage Drop).
- **Current Source Constraint Equation:**
  - Look at the shared branch containing the current source I_s.
  - Whichever mesh current flows in the **same direction** as the current source arrow gets a **positive (+)** sign.
  - Whichever mesh current flows in the **opposite direction** gets a **negative (-)** sign.

$$\text{Current in arrow's direction} - \text{Current against arrow's direction} = I_s$$

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3. Step-by-Step Supermesh Procedure

Follow this algorithmic process for any circuit containing a current source shared by two meshes:

1. **Assign Mesh Currents:** Assign clockwise mesh currents $(I_1, I_2, \ldots, I_n)$ to all meshes in the circuit.
2. **Identify Shared Current Sources:** Locate any current source shared between two meshes.
3. **Form the Supermesh:** Remove the shared current source branch mentally or visually to form a single larger outer loop encompassing both meshes.
4. **Write the Supermesh KVL Equation:** Traverse the perimeter of the combined supermesh in a clockwise direction, summing all voltage drops and gains to zero.
5. **Write the Constraint Equation:** Express the value of the shared current source in terms of the defined mesh currents.
6. **Solve the System of Equations:** Use substitution or matrix elimination (Cramer's Rule) to solve for the unknown currents.

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4. Worked Example & Problem Solution

Circuit Diagram

Diagram: Supermesh Circuit with Two Meshes sharing a 2A Current Source

> **Visual Description of the Circuit:**
>
> - **Left Branch:** Contains a $10\ \text{V}$ DC independent voltage source with the positive terminal on top and negative terminal on the bottom, in series with a $4\,\Omega$ resistor along the top rail.
> - **Middle Branch (Shared):** Contains an independent current source of $2\ \text{A}$ pointing vertically **upward**.
> - **Right Top Rail:** Contains a $6\,\Omega$ resistor.
> - **Right Branch:** Contains a $6\ \text{V}$ DC independent voltage source with the positive terminal on top and negative terminal on the bottom.
> - **Bottom Right Rail:** Contains a $2\,\Omega$ resistor.
> - **Mesh Currents:** Mesh 1 (left loop) has current I_1 flowing clockwise. Mesh 2 (right loop) has current I_2 flowing clockwise.

Step-by-Step Detailed Solution

Step 1: Define Mesh Currents

- Let I_1 be the clockwise mesh current in the left loop (Mesh 1).
- Let I_2 be the clockwise mesh current in the right loop (Mesh 2).

Step 2: Identify the Supermesh and Write the Constraint Equation

The $2\ \mathrm{A}$ current source is shared between Mesh 1 and Mesh 2.

- In the middle branch, I_1 flows **downward**.
- In the middle branch, I_2 flows **upward**.
- The $2\ \mathrm{A}$ source is pointing **upward**.

Since I_2 flows in the same direction as the $2\ \mathrm{A}$ current source and I_1 flows against it:

$$I_2 - I_1 = 2$$

Re-arranging to express I_1 in terms of I_2:

$$I_1 = I_2 - 2 \quad \text{— (Equation 1)}$$

Step 3: Form the Supermesh and Apply KVL

By temporarily removing the central $2\ \mathrm{A}$ branch, we form a single combined Supermesh surrounding both Mesh 1 and Mesh 2.

We traverse the outer loop of the Supermesh in a clockwise direction, starting from the bottom-left corner near the $10\ \mathrm{V}$ source:

1. **Through the $10\ \mathrm{V}$ Source (bottom to top):**
   Moving from $-$ to $+$, this is a voltage gain:
   $$+10$$
2. **Through the $4\,\Omega$ Resistor (left to right):**
   This resistor lies exclusively in Mesh 1, so the current flowing through it is I_1:
   $$-4 \cdot I_1$$
3. **Through the $6\,\Omega$ Resistor (left to right):**
   This resistor lies exclusively in Mesh 2, so the current flowing through it is I_2:
   $$-6 \cdot I_2$$
4. **Through the $6\ \mathrm{V}$ Source (top to bottom):**
   Moving from $+$ to $-$, this is a voltage drop:
   $$-6$$
5. **Through the $2\,\Omega$ Resistor (right to left):**
   This resistor lies exclusively in Mesh 2, so the current flowing through it is I_2:
   $$-2 \cdot I_2$$

Summing all terms according to KVL:

$$10 - 4I_1 - 6I_2 - 6 - 2I_2 = 0$$

Step 4: Simplify the Supermesh KVL Equation

Combine like terms in the KVL equation:

$$(10 - 6) - 4I_1 - (6I_2 + 2I_2) = 0$$

$$4 - 4I_1 - 8I_2 = 0$$

Divide the entire equation by 4:

$$1 - I_1 - 2I_2 = 0$$

$$I_1 + 2I_2 = 1 \quad \text{— (Equation 2)}$$

Step 5: Solve the System of Simultaneous Equations

Substitute **Equation 1** ($I_1 = I_2 - 2$) into **Equation 2**:

$$(I_2 - 2) + 2I_2 = 1$$

$$3I_2 - 2 = 1$$

$$3I_2 = 3$$

$$I_2 = 1 \text{ A}$$

Now, substitute $I_2 = 1 \text{ A}$ back into **Equation 1** to solve for I_1:

$$I_1 = 1 - 2 = -1 \text{ A}$$

Final Calculated Values

- **Mesh Current I_1:** -1 A *(The negative sign simply indicates that actual current I_1 flows counter-clockwise)*
- **Mesh Current I_2:** 1 A

$$\mathbf{I_2 = 1 \text{ A}}$$